The great tit HapMap project: a continental-scale analysis of genomic variation in a songbird

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Abstract

A major aim of evolutionary biology is to understand why patterns of genomic variation among populations and species. Widespread continental species are interesting models for studying how evolution shapes patterns of genomic divergence, as they are associated with recent and rapid colonisations, large population sizes and high levels of gene flow. We studied patterns of genomic variation among 28 great tit (*Parus major*) populations in Europe. We found that genome-wide variation was consistent with a recent colonisation across Europe from a single refugium in the Balkans and/or Turkey. The highest levels of genome-wide structure were found between island populations in Crete, Sardinia and Corsica, most likely as a result of bottlenecks and reduced gene flow among these populations. We then found that differentiation across the genome was highly heterogeneous, with clear "islands of differentiation", even among the most recently separated populations. These islands of differentiation were consistently found in regions of low recombination, suggesting that background selection can rapidly promote population differentiation among recently colonised populations. We also detected outlier regions that were unique to peripheral great tit populations, most likely as a result of recent directional selection at the range edges of this species. These "unique" outlier regions contained candidate genes for morphology, stress response and colouration, which both support previous research in this species and provide avenues for future investigation. Together, our results suggest that examining differentiation landscapes in the context of population history is a useful way for determining how natural selection shapes genomic variation in the wild.

Introduction

Since the first studies of allozyme variation in humans [1] and *Drosophila* [2,3], there has been great interest in explaining how evolutionary and ecological processes shape the patterns of genetic variation observed within and among natural populations. One focus of research and debate in this area has been on quantifying the roles of adaptive and neutral processes in explaining observed levels of genetic variation [4]. However, adaptation does not occur in isolation, but against a background of recombination, gene flow, and genetic drift. More recently there has been increased effort in understanding how these fundamental evolutionary forces operate in concert to generate and maintain the levels of genetic diversity commonly observed in natural populations [5,6].

The arrival of high-throughput sequencing and subsequent characterisation of genome-wide variation across large numbers of individuals has revealed that at the genetic level, patterns of variation and divergence among natural populations and species are highly heterogeneous [7]. A feature of these "genomic landscapes"

of divergence that has received particular attention is the presence of so-called "islands of differentiation"; outlier regions of the genome with high levels of divergence estimated from statistics such as F_{ST} or d_{xy} [7–10]. Initially these regions were termed "islands of speciation", and were thought to arise as a result of reduced gene flow in genomic regions associated with reproductive isolation [7,11]. Subsequent modelling and analysis has revealed that highly heterogeneous patterns of genomic divergence can occur in the complete absence of gene flow, as a result of recombination rate variation and linked selection [12,13]. In genomic regions of low recombination, selection for beneficial mutations (positive selection), or against deleterious mutations (background selection), will impact relatively large genomic regions as a result of high levels of linkage disequilibrium (LD) among sites. Selection within these regions reduces diversity within populations, and increases differentiation among them, resulting in "islands" of increased differentation that persist over evolutionary time [**???**,13]

Comparing patterns of genomic differentiation among sets of populations and/or species with different different levels of genome-wide divergence is a powerful way of disentangling the forces that shape variation among populations. Martin et al. [14] showed that, across multiple *Heliconius* butterfly populations and species, patterns of genomic variation were shaped by a combination of gene flow and selection, particularly in genomic regions harbouring genes involved in wing patterning. In contrast, Renaut et al. [9] showed that in *Helianthus* sunflowers, genomic architecture was the main driver of genomic differentiation across sets of populations. Similarly, recent research in birds has revealed that differentiation landscapes are conserved across populations, species and even across avian families, with the same islands of differentiation arising among populations of distantly related species [15–17]. This latter pattern appears to have arisen, at least in part, as a result of a highly conserved recombination landscape in birds, with background selection in regions of low recombination producing recurrent islands of differentiation [18].

It is now clear that the recombination landscape and linked selection are key drivers of genomic variation within and among populations. However, we are only just beginning to understand how this linked selection interacts with other evolutionary forces to shape patterns of differentiation across natural populations and species [18–22]. A recent, large-scale analysis in sticklebacks showed that outlier genomic regions were more likely to arise in low recombination regions when gene flow occurred between populations [23]. There is also a significant impact of divergence time; in very recently separated populations the differentiation landscape is most likely to reflect selective sweeps, then as divergence accumulates, genomic architecture is expected to play an increasingly important role [18].

Widespread continental species are excellent models for studying how demography and the environment shape genetic and phenotypic variation among populations, due to their large effective population sizes and

ecologically varied ranges. Patterns of genome-wide variation in continental species is typically shaped by post-glacial colonisation routes and contemporary gene flow [24,25], although local adaptation can also play a role in shaping patterns of genetic diversity [**???**]. Genomic studies of continental species have enabled accurate assessments of colobisation routes, and revealed adaptation to environmental conditions REFS. However, to our knowledge, no studies have examined how the landscape of genomic differentiation changes in the context of postglacial recolonisation of continental environments. Large-scale studies of genomic differentiation across widespread species may inform how divergence accumulates in low recombining regions among recently separated populations, as well as how selection, gene flow and genomic architecture interact to shape genomic divergence in the early stages of population separation.

The great tit (*Parus major*) is a widespread passerine and widely used model for ecological and evolutionary study [26]. Phyleogeographic research using mitochondrial and microsatellite loci suggests that this species has recently expanded across its European range, possibly from a single refugium in the Balkans [27]. Contemporary populations are characterised by large effective population sizes and high levels of gene flow among populations, resulting in low levels of genetic differentiation [28,29]. The genome of the great tit has recently been sequenced [30], and a high density (650K) panel of SNP markers has been developed. A recent study of two European populations using this marker panel suggests that rapid adaptation has occurred at the genomic and phenotypic levels [31]. A large number of studies containing a wealth of ecological data have been carried out across multiple great tit populations [32–34], enabling us to generate informed hypotheses about selection in this system.

In this study we examined how genomic architecture, natural selection and population history have shaped patterns of genomic variation across recently colonised European great tit populations. Using a large SNP panel typed across 28 populations (Fig. 1), we first characterise genome-wide patterns of variation within and among populations, in order to infer population history. We then examine how variation is partitioned across the genome, and test the hypothesis that highly divergent genomic regions have arisen in regions of low recombination [12,13]. Finally, we examine how genomic divergence accumulates along the colonisation route of this species, with the aim of inferring how recent natural selection and background selection drive variation across the genome in the wild.

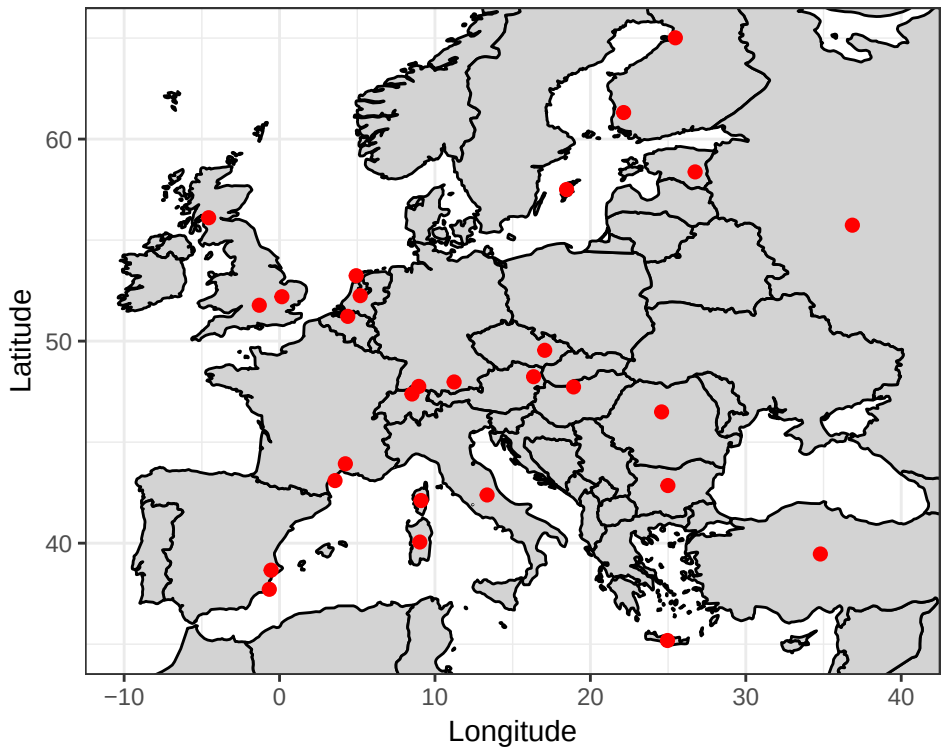

Figure 1 Sampling locations of great tit populations used in the present study.

Results and Discussion

Genetic diversity and population history

Sampling locations and sample sizes for each population are given in Table 1. Levels of genetic diversity (π_{SNP}) were generally high, but we observed substantial differences among populations (Fig. 2A). Similarly, LD declined rapidly in all populations, reaching a baseline levels within ~5kb in all populations, but also varying among populations (Fig. S1). Highest levels of LD (and lowest levels of genetic diversity) were observed in the island populations of Crete (Greece), Sardinia (Italy), Corsica (France) and Vlieland (Netherlands) with lowest levels of LD in central Europe, Turkey and Spain (Fig. 2A, Fig. S1). This is consistent with reduced effective population size and reduced gene flow in these populations, either as result of the colonisation process, or more recent bottlenecks [35].

Genome-wide F_{ST} between European great tit populations was 0.007, with no significant pattern of isolation-by-distance (Mantel test; r = 0.14, p = 0.16). Instead, the highest levels of F_{ST} were found in comparisons involving the island populations of Corsica, Sardinia and Crete (Fig. S2). Admixture analysis was consistent with this pattern (Fig. S3); the $K = 2$ analysis assigned individuals in Sardinia and Corsica to one genetic

cluster, and the remaining populations to the second. Thus, it is likely that much of the genetic structure between European great tit populations is a result of genetic drift in small island populations. Admixture also revealed some structure between (mainly peripheral) mainland and larger island populations. At $K = 3$ (the model that best fitted the great tit data; Fig. S3), Spain was separated from the rest of mainland Europe. Increasing values of K resulted of the separation of populations in Scotland ($K = 4$), Sardinia (from Corsica; $K = 5$), France ($K = 6$), Crete ($K = 7$) and England ($K = 8$). The Admixture output at $K = 8$ is displayed in Fig. 2C; further increases in K did not generate patterns of structure that corresponded to geographical variation (Fig. S3), and were increasingly less well supported (Fig. S4). Thus, even with hundreds of thousands of markers Admixture was unable to separate many of the central European populations, confirming that levels of divergence are extremely low. PCA largely corroborated the Admixture results, with PC1 separating Corsical and Sardinia from the remaining populations, PC2 separating Spain, PC3 separating Corsical from Sardinina, and PC4 separating Scotland, England and France (Fig. S5).

Maximum likelihood analyses implemented in TreeMix showed that a model with no migration explained 97.8% of variance in relatedness between populations; increasing the number of migration events substantially improved the percentage of relatedness explained, up to 99.7% when 10 migration events were fitted (Fig. S6). In Figure 3 we display the maximum likelihood tree with two migration events, after which the variance in relatedness explained plateaued when more migration events were added (Fig. S6). The tree was generally characterised by short branch lengths, with the exception of the island populations of Sardinia and Crete, which were grouped with the population from mainland Italy. Thus, the TreeMix analysis is consistent with the pattern of low overall genomic divergence, with the exception of island poppulations. However, much of the grouping that did occur among continental populations made geographical sense, with populations from Finland and Estonia grouped together, as were populations from Turkey and the Balkans, and populations from England and Scotland (Fig. 3). Interestingly, TreeMix grouped the Spanish and Corsican populations, which is consistent with previous subspecies descriptions of European great tits [36]. The two fitted migration edges both involved Sardinia, involving migration from eastern Europe, and to Corsica (Fig. 3).

We tested the hypothesis that European great tits colonised from a single refugium in Turkey and the Balkans [27] by comparing genetic and geographic distance between each population and the proposed refugial populations. Becasuse of the elevated structure in Corsica, Sardinia and Crete (Fig. S2), we excluded comparisons involving these populations. We found that F_{ST} was significantly related to distance from Turkey ($r = 0.84$, $p < 0.001$; Fig. 2C) and the Balkans ($r = 0.47$, $p < 0.001$). The same relationship was not found for alternative potential refugial populations in Spain ($r = -0.07$, $p = 0.65$), or Italy ($r = 0.15$, $p = 0.49$). Our results therefore lend empirical support to the hypothesis [27] that great tits colonised Europe

from a single refugium in the southeast.

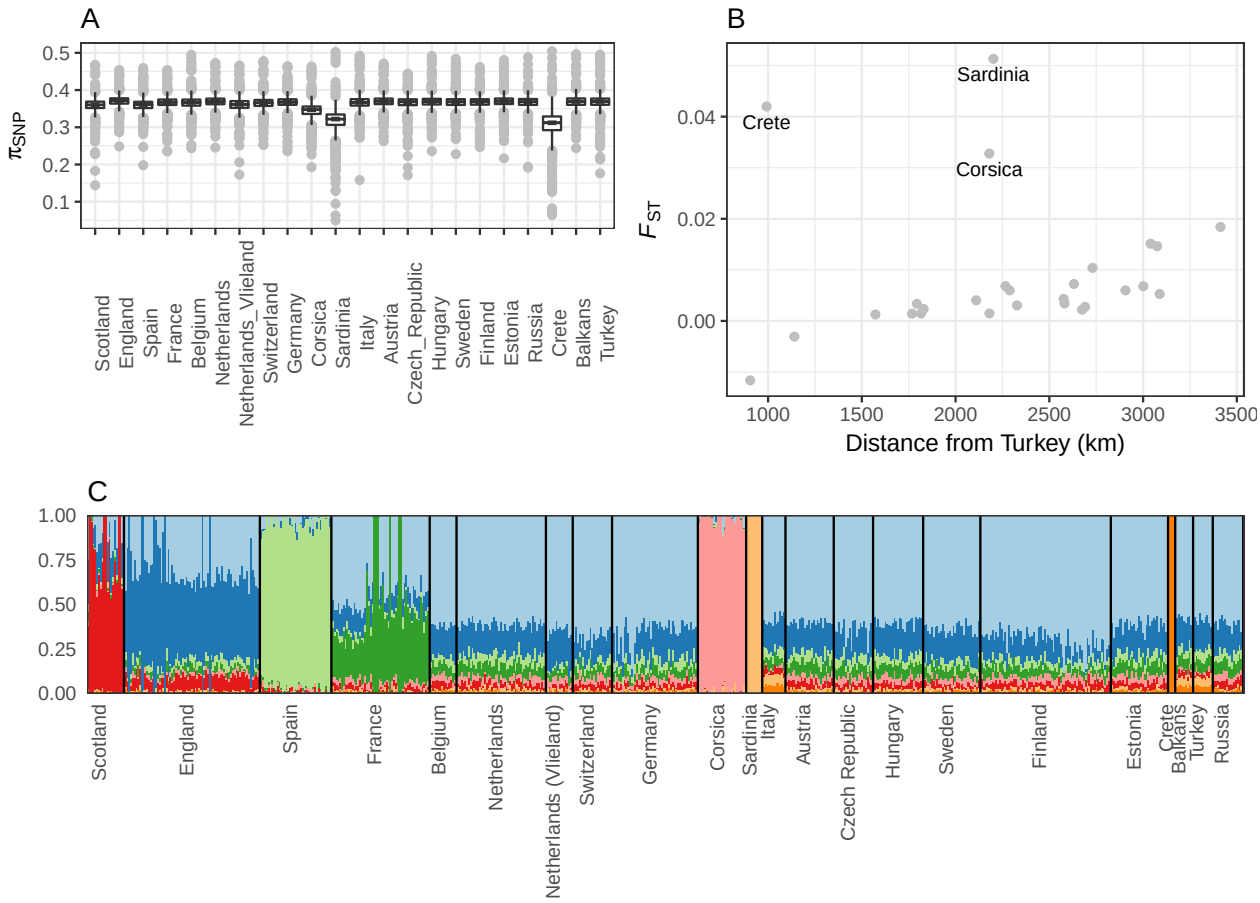

Figure 2 Genetic diversity and structure in European Great tit populations. **A** Nucleotide diversity within each population. **B** Pairwise F_{ST} in relation to geographic distance from the Turkey, only including comparisons involving Turkey. **C** Output from admixture analysis at $K = 8$.

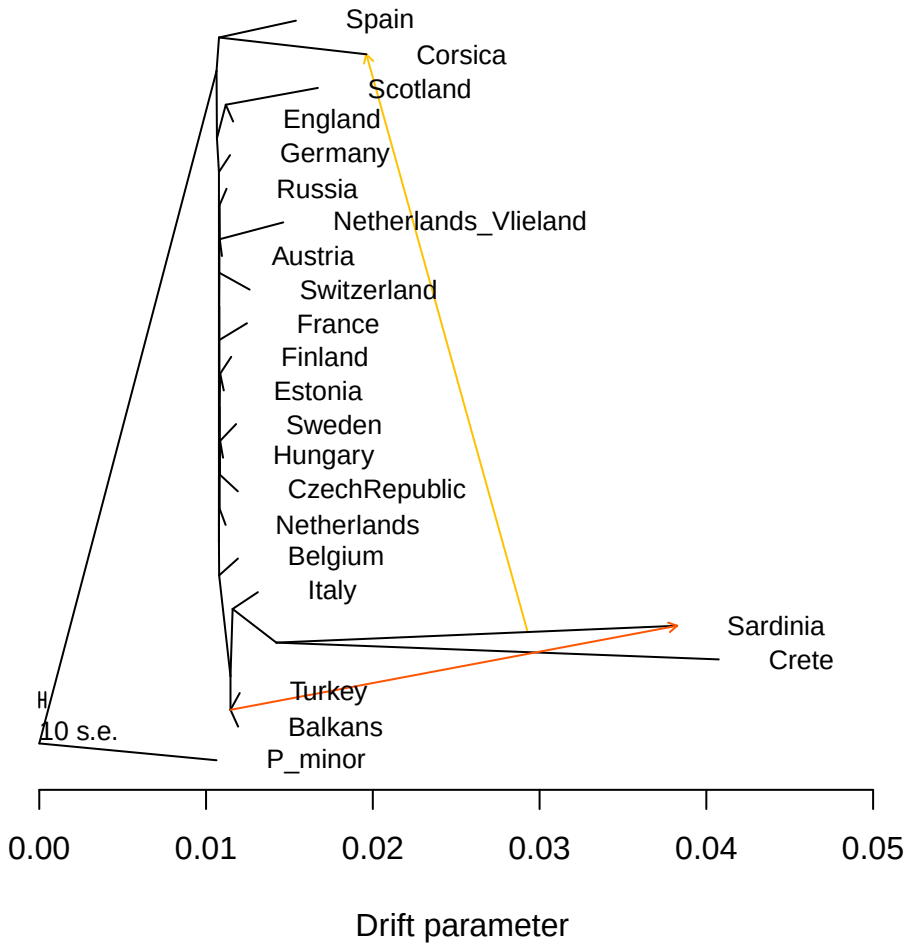

Figure 3 Maximum likelihood tree inferred by TreeMix, allowing ten migration events. Migrations events (arrows) are coloured according to their weight (red = higher migration), and horizontal branch lengths are proportional to the amount of genetic drift that has occurred along the branch.

Genomic landscapes of differentiation

It is likely that the many, and perhaps the majority, of wild populations are characterised by highly heterogeneous patterns of differentiation across the genome [22]. To examine how landscapes of genomic divergence have formed along the colonisation route of European great tits, we calculated windowed F_{ST} in 500kb bins between each population and the proposed refugial population in Turkey. We found that F_{ST} varied markedly across the genome in all comparisons (Fig. 4; Fig. S7). Outlier regions (windows with $zF_{ST} > 10$, where zF_{ST} is F_{ST} expressed relative to the genome-wide average, in standard deviations) were found in all comparisons apart from Crete and Sardinia, where overall levels of divergence were highest (Fig. 4). Our results suggest, therefore, that islands of differentiation arise even among populations that could be considered panmictic.

Genomic differentiation with Turkey was negatively related to recombination rate in almost all populations (Fig. 5). The relationship between F_{ST} and recombination rare was generally weak, with correlation coefficients < 0.1. In a handful of populations this relationship was substantially stronger - most notably in the island populations of Corsica, Sardinia and Crete, and in England, Scotland and Spain (Fig. 5). Outlier regions of very high differentiation almost exclusively occurred in relatively low recombining regions, and accordingly: the recombination rate of outlier regions was significantly lower than the genome-wide average (Wilcoxon test, $P < 0.001$), suggesting that linked selection in regions of low recombination plays a key role in driving patterns of genomic differentiation even in the very earliest stages of population separation.

Of the 11 outlier regions, nine were found in only one or two comparisons, while the other two were found in 12 and 10 comparisons, respectively (Table S2). We hereafter refer to outlier regions found in one or two comparisons as "unique" outlier regions, and to those found in more than two comparisons as "shared" regions. Unique regions tended to be found in the most peripheral European great tits, with three found in Scotland, two in England, Spain and Finland; the remaining outlier regions were found in comparisons involving the Czech republic, Russia, Vlieland (Netherlands) and Belgium (Table S2). There appeared to be no difference in the recombination rate between shared and unique outlier regions (Fig. 5), although the small number of regions precluded testing this hypothesis formally.

It is difficult to differentiate between positive and background selection in producing heterogeneous patterns of differentiation across the genome. Correlated patterns of genomic differentiation among independent sets of populations suggests that background selection likely plays a role [15–17]. It appears that European great tits have colonised from a single refugial population, and as such truly independent comparisons are not available. Therefore an outlier region shared among multiple populations could represent either selection in

the ancestral population, or background selection. Given the pervasiveness of background selection in birds [16,17], and other organisms [37], and the overall negative relationship between F_{ST} and recombination rate in this study (Fig. 5), it is likely that this background selection is the driver of the shared outlier regions in great tits. Thus, it appears that background selection can generate islands of differentiation in the very earliest stages of population separation. This is not necessarily what we would expect - it is often assumed that peaks of high differentiation in recently separated populations are the product of selective sweeps, and that only after time do correlated patterns of genomic divergence arise [18,23]. Further research into genomic landscapes of differentiation among widespread continental species will shed further light on the role of background selection in shaping genomic divergence over short evolutionary timescales.

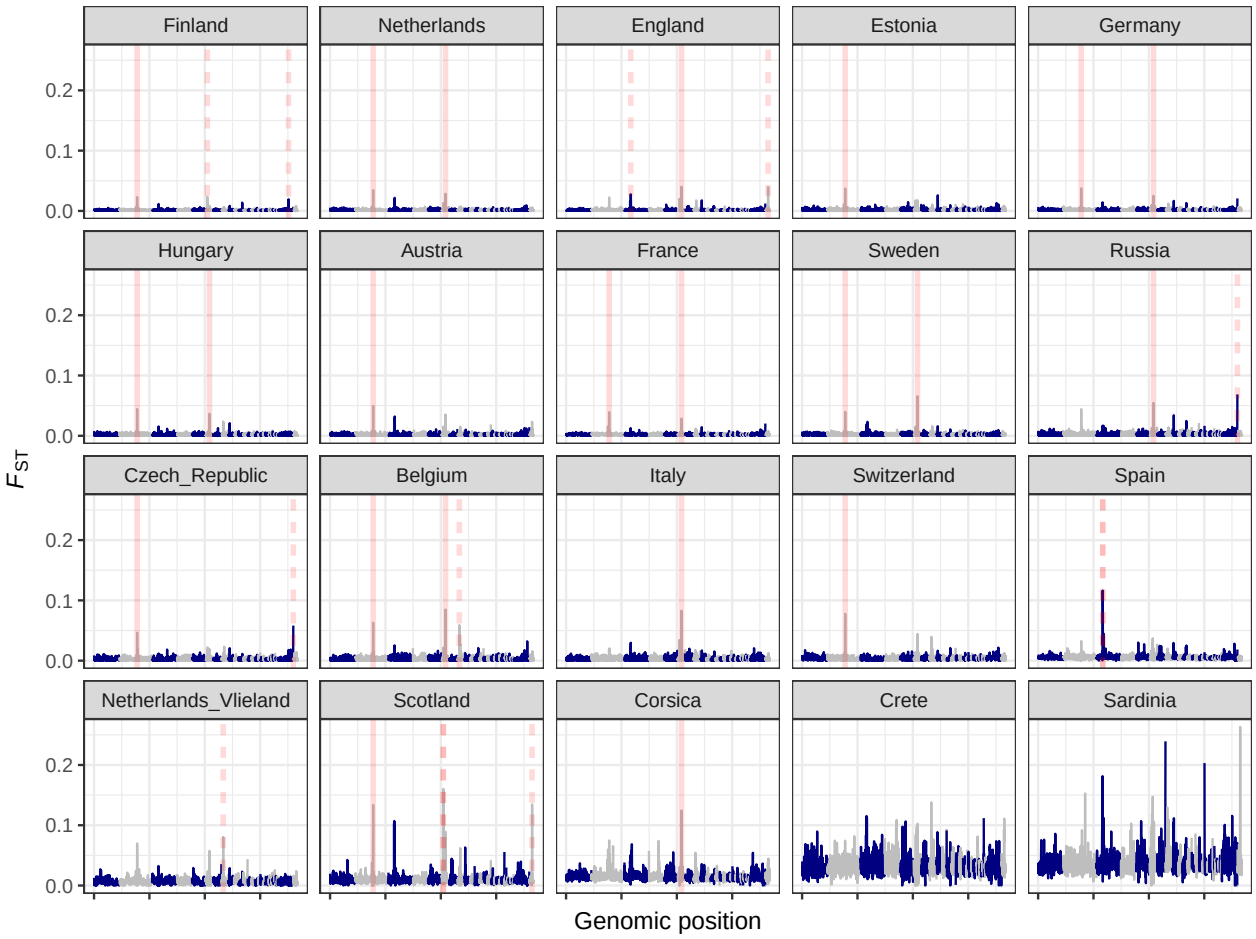

Figure 4 Landscapes of genomic differentiation in European great tit populations. F_{ST} across the genome is averaged in 500kb windows, with each panel displaying a pairwise comparison between with the proposed refugial population in Turkey. Red lines represent F_{ST} outliers (windows with mean F_{ST} values at least 10 standard deviations greater than the global mean for that comparison) shared across more than two comparisons (solid red lines), or unique to one or two comparisons (dashed red lines).

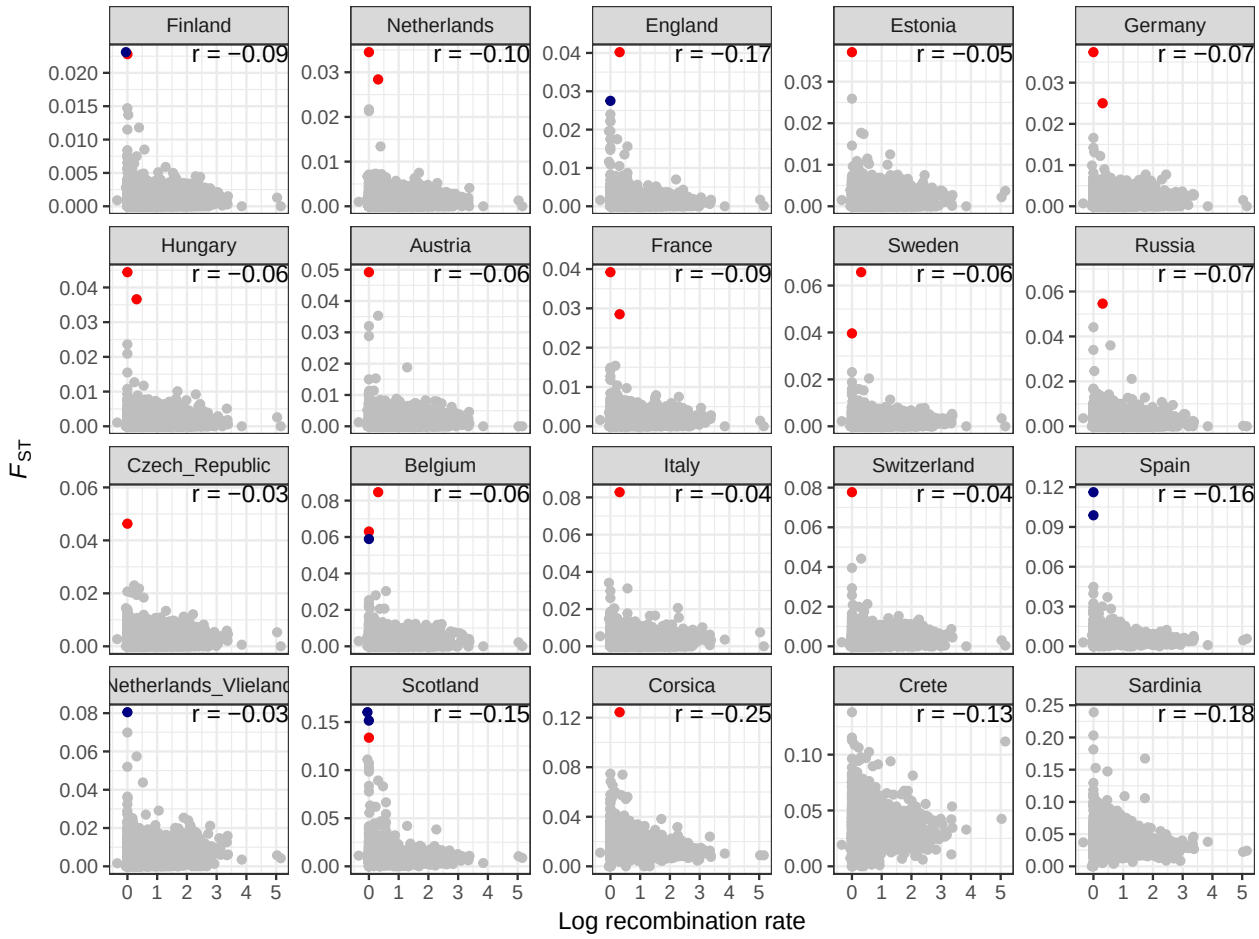

Figure 5 Genomic differentiation and recombination rate variation in European great tit populations. Each point is a 500kb window, with each panel displaying a pairwise comparison between with the proposed refugial population in Turkey. Coloured points represent F_{ST} outliers (mean zF_{ST} values > 10) shared across more than two comparisons (red), or unique to one or two comparisons (dark blue).

To further explore how selection may have shaped variation in F_{ST} outlier regions, we estimated levels of nucleotide and haplotype diversity within these regions. Nucleotide diversity (π_{SNP}) in outlier regions varied from 0.28 to 0.41, and diversity in these regions was significantly lower than the genome-wide average (Wilcoxon test, $P = 0.007$; Fig. 6A). However, there appeared to be no difference in nucleotide diversity between shared and unique regions (Fig. 6A). Haplotype diversity varied substantially among regions, with haplotype richness ranging from 72 to 1033. Both haplotype richness and marker density in shared regions tended to be lower than those in unique regions (Fig. 6B). A detailed examination of haplotype structure in these regions is displayed in Figure S8. Although more distinct regions are needed to properly test underlying differences in haplotype structure, we provide a detailed examination of haplotype structure in two regions typically containing a signal in 1 or multiple regions (Fig S7). The frequency of a single haplotype was much higher in Oulu than in other populations, indicating a population-specific selective sweep. However, the region on chromosome 2 containing a signal in multiple populations is much less complex, demonstrating higher haplotype frequencies across a multitude of populations.

Our data suggest that examining patterns of nucleotide and haplotype diversity in outlier regions may help to separate recent episodes of positive selection from the gradual process of background selection (Figs 6, S7). Population comparisons allow for the identification of haplotypes at high frequency in single populations, a patterns that is unexpected by chance giving the high background in haplotype diversity in other populations. By contrast, the genomic architecture of regions that contain a high level of differentiation in multiple populations indeed seems to be less complex. We stress that marker density, haplotype richness and the frequency of the most common haplotype provide important insight into the efficiency of selection. Comparing these statistics in multiple populations enable the distinction between recent positive selection and drift effects. However, as we only detected a small number of outlier regions we are unable to make a detailed comparison between shared and unique regions. Further, simulations suggest that haplotype-frequency methods for detecting selection can be sensitive to background selection [38]. Research which combines multiple populations to allow shared and unique outlier regions to be identified, alongside a range of haplotype and SNP based methods for detecting selection, may allow us to identify the roles of positive and background selection in shaping variation.

Genes found in the outlier regions are displayed in Table S2. Perhaps most notable among these is COL4A5, a gene found to be associated with bill length, and under selection between populations between England and the Netherlands, in a recent great tit study [31]. Here we found that the COL4A5 region is an F_{ST} outlier in England and Scotland, but not in any other European populations (Table S2). Another notable candidate gene potentially involved in beak morphology, and previously found to be under selection in UK

great tits is BMPR1A, which in this study was found in an outlier region in Scotland. Other genes in the unique outlier regions included potential candidates for morphology and obesity (PPP1CB [39], GHITM [40]), thermal stress response (CDKN1B [41]), malaria infection (MRPL33 [42]) and colour variation (SOX10 [43]).

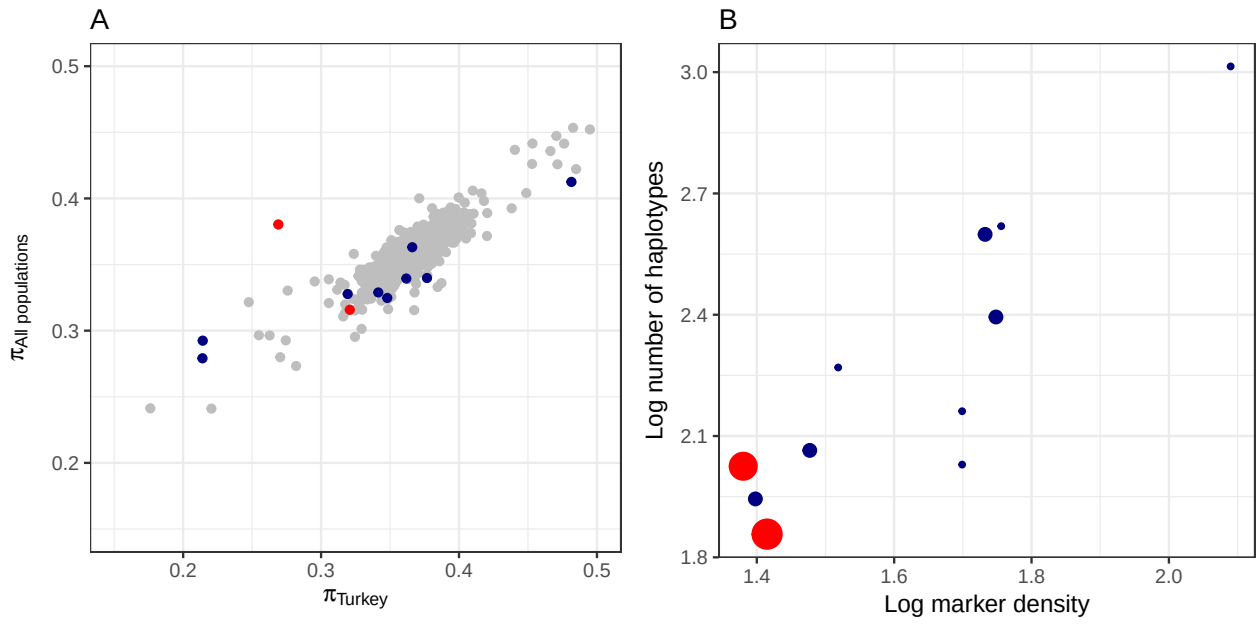

Figure 6 Genetic diversity in F_{ST} outlier regions in European great tit populations. Each point is an outlier window, with point size scaled by the number of populations the outlier was found in - "shared" haplotypes are coloured red and "unique" haplotypes dark blue (see main text for details). In **A**, Nucleotide diversity for each window is calculated for the proposed refugial population in Turkey, and plotted against the mean for all other populations. In **B**, SNP marker density is plotted against haplotype richness for each region.

Regions of high differentiation that are not shared among populations are more likely to be the result of recent positive selection [18]. We found evidence for such non-shared regions in European great tits, and that these tended to occur in peripheral populations. Observational and experimental research shows that adaptation at range edges is a key feature shaping divergence among recently colonised and expanding populations [44–46]. In great tits, previous research has shown evidence for adaptation in marginal populations. A comparison of Swedish and Finnish populations revealed genetic differences in basal metabolic rate, most likely an adaptive response to winter conditions [47]. Here we found outlier regions unique to populations in cold populations in Scotland, Finland and Russia, containing at least one candidate gene for thermal stress [41]. The detection of an outlier region containing the gene COL4A5 was particularly interesting, as this region was detected in a previous comparison of UK and Dutch great tits [31], in which significant associations with bill length and fitness were detected. Our results here are in accordance with the description of UK great tits as a separate subspecies [48], and combined with previous results suggest that this divergence is the result of recent natural selection in the UK [31]. Numerous other potential candidate genes were detected in these marginal populations, with potential roles in morphology, colouration, obesity and malaria infection. We are reluctant to speculate whether these candidate genes are genuine targets for natural selection, and more

reluctant still to speculate as to how selection might be driving variation at these region. Regardless, these candidates will provide useful starting points for future genomic and ecological investigation.

Finally, it is worth noting that environmental association methods are highly suited method for detecting adaptation in widespread continental species [49,50], and further work will test how variation in the abiotic environment has shaped patterns of genomic variation in great tits. To our knowledge there has been little research on the potential for background selection to generate false positives in environmental association analyses, and given the pervasiveness of background selection, this should now be a priority. However, careful association of environmental and genotype data in species such as great tits, in which a wealth of ecological and genomic resources are available, is likely to generate interesting insights into the the genetic and phenotypic basis of natural selection.

Materials and Methods

Sampling and molecular methods

Samples were collected from 28 populations from 22 countries across Europe (Fig. 1; Table S1). Birds were trapped from nest boxes, or using mist nets, and ringed with a uniquely numbered aluminium ring. Blood was taken via brachial venipuncture, and stored in either 1 ml Cell Lysis Solution (Gentra Puregene Kit, Qiagen, USA) or Queen's buffer. All samples were genotyped using a custom-made Affymetrix® great tit 650K SNP chip at Edinburgh Genomics (Edinburgh, United Kingdom). After filtering, a total of 647 samples typed at 483888 SNPs were retained for analysis.

Analyses

Unless stated otherwise, all population genetic statistics were calculated in PLINK version 1.9 [51], and downstream analysis and plotting was carried out in R version 3.3 [52]. In each population, we estimated LD (R^2) for each pair of markers within 50kb on the same chromosome, and compared this to physical distance between marker pairs. For a simple comparison of short-range LD among populations, we also calculated, for each population, mean LD between all pairs of markers within 1kb. We calculated observed heterozygosity for each SNP and population using a reduced SNP dataset, which was pruned based on LD to remove all markers with $R^2 > 0.1$, then thinned with a probability of retaining each variant of 0.25.

We calculated genome-wide (mean) F_{ST} between each pair of populations using the pruned and thinned dataset described above. Pairwise F_{ST} was compared to geographic distance between populations using Mantel tests, implemented in the Ecodist package in R [53]. We tested whether genetic structure was related

to distance from candidate refugial populations (in the Balkans, Turkey, Spain and Italy), using Pearson correlations. We also estimated population structure using Admixture version 1.3, with default settings [54]. We varied values of K from one to ten; by which point increasing values of K provided no informative information about population structure (see results). Model support for each value of K was estimated by calculating 5-fold cross-validation error. Finally, we visualised the evolutionary history among European great populations by generating a maximum likelihood tree in TreeMix version 1.13 [55]. We rooted the tree using a sample of *P. minor* individuals sampled from Amur, Russia. We fitted models allowing for range of migration events (0-10), and used a window size of 500 SNPs [55]. To assess model fit, we calculated the proportion of variance in relatedness between populations explained by each model [55].

We examined the genomic landscape of differentiation across European great tit populations by calculating F_{ST} in 500kb bins, using python scripts obtained from Github (https://github.com/simonhmartin/genomics_general). We did not estimate d_{xy}, as this parameter is difficult to estimate accurately from single SNP loci [13]. We also calculated standardised F_{ST} (hereafter ZF_{ST}) by mean-centring windowed values and dividing them by the standard deviation among windows. We defined outlier regions as 500kb bins with zF_{ST} values greater than ten. We tested whether the landscape of genomic differentiation was related to recombination rate variation using a recombination map previously developed for the great tit using a 10K SNP chip [56], from which we estimated recombination rates in 500kb non-overlapping windows.

Acknowledgements

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